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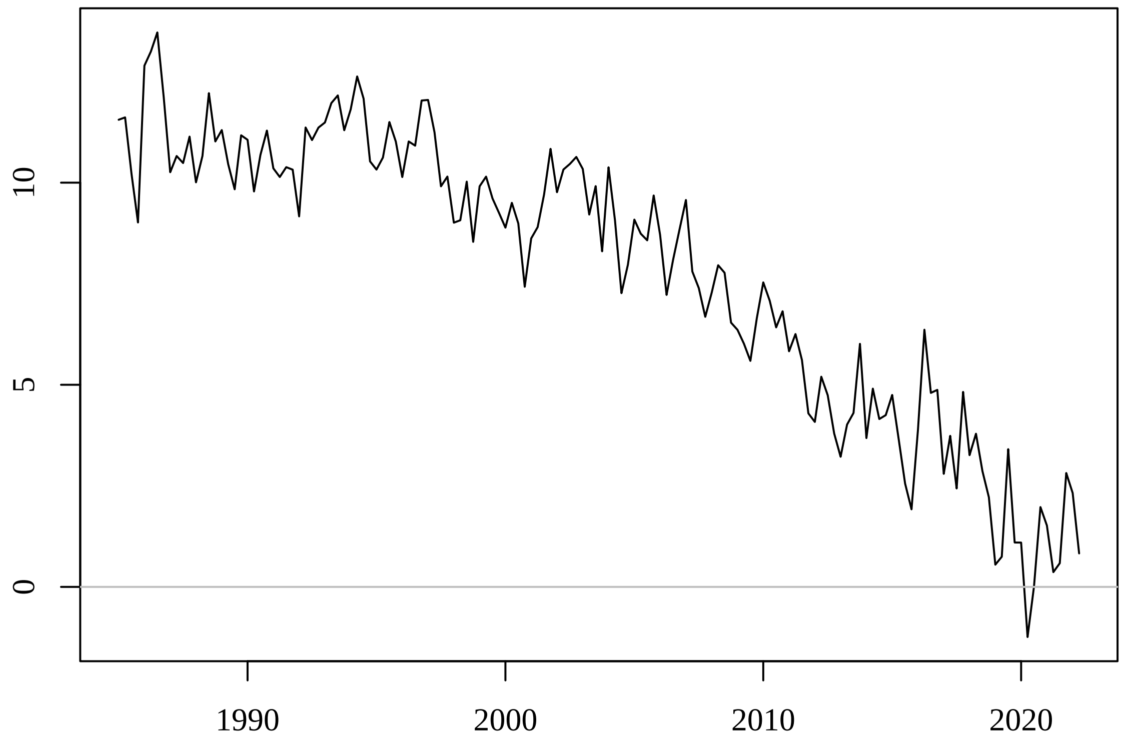
Question 1. (8 marks)

The following is a plot of a quarterly time series from 1985q1 to 2022q2, a total of 150 observations.



(a) **(2 marks)** Does this time series plot suggest a **trend** is present? Answer "yes", "no" or "can't tell", with a one sentence explanation.

(b) **(2 marks)** Does the time series plot suggest a **trend break** is present? Answer “yes”, “no” or “can’t tell”, with a one sentence explanation.

(c) **(2 marks)** Does the time series plot suggest **non-stationarity** is present? Answer “yes”, “no” or “can’t tell”, with a one sentence explanation.

(d) **(2 marks)** Does the time series plot suggest **non-invertibility** is present? Answer “yes”, “no” or “can’t tell”, with a one sentence explanation.

Question 2. (15 marks)

An augmented Dickey-Fuller test regression with trend was calculated for the time series in the plot in Question 1 using the `urca` command `ur.df`:

```
ur.df(Y, type="trend", lags=1)
```

The following test regression was obtained.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	3.860	0.895	4.314	0.000
z.lag.1	-0.289	0.066	-4.413	0.000
tt	-0.022	0.005	-4.264	0.000
z.diff.lag	-0.085	0.083	-1.028	0.306

(a) **(2 marks)** Write out the estimated test regression in equation form.

(b) **(1 mark)** What is the value of the augmented Dickey-Fuller t test statistic?

(c) **(2 marks)** What are the null and alternative hypotheses of the Dickey-Fuller test?

The `urca` command `qunitroot(q, trend="ct")` is used to compute the critical values for the augmented Dickey-Fuller test for `q = 0.05, 0.01, 0.005, 0.001`. The output is shown here:

```
                        5%     1%    0.5%   0.1%
ADF critical value: -3.41 -3.958 -4.164 -4.596
```

(d) **(3 marks)** Carry out the augmented Dickey-Fuller test using the statistic in part (b) using the 5% level of significance. State the test decision and conclusion.

(e) **(3 marks)** Suppose we decide there is a break in the trend line occuring in quarter 1 of 2003, which is observation $t = 80$, such that the trend line remains continuous at that point. Write out the form of the test equation that would be estimated to include this trend break. Clearly define any additional notation you need.

(f) **(5 marks)** Outline the steps involved in computing an appropriate 5% critical value for the test in part (e).

If you run out of space for Question 2, use the extra pages at the end of the exam paper. Please clearly indicate there the question number/part you are answering.

☐ **Please tick this box if you used extra pages for any part of Question 2.**

Question 3. (23 marks)

For a stationary time series with $n = 200$ observations a range of ARMA(p,q) were fitted to attempt to find a suitable forecasting model. The following table shows the values of the AICc and p-values of the Ljung-Box residual autocorrelation test for every combination of $p = 0, 1, 2, 3, 4$ and $q = 0, 1, 2$.

p	q	AICc	LBp
0	0	1.070	0.000
1	0	1.016	0.048
2	0	1.011	0.148
3	0	1.005	0.872
4	0	1.008	0.311
0	1	1.000	0.835
1	1	1.003	0.800
2	1	1.007	0.647
3	1	1.008	0.881
4	1	1.009	0.287
0	2	1.003	0.841
1	2	1.007	0.796
2	2	1.010	0.764
3	2	1.011	0.909
4	2	1.013	0.292

(a) **(4 marks)** Based on the Ljung-Box p values (`LBp`), which model(s) may be considered to be **mis**specified? Justify your conclusions.

(b) (3 marks) According to the AICc alone, what is the best model? Is there evidence this model is misspecified? Explain.

(c) (3 marks) According to the AICc alone, what is the best AR(p) model? Is there evidence this model is misspecified? Explain.

The coefficient estimates of the ARMA(0,1) and ARMA(3,0) models are as follows:

```
                ma1 intercept
ARMA(0,1): -0.491     3.014

                ar1    ar2    ar3 intercept
ARMA(3,0): -0.476 -0.234 -0.166     3.014
```

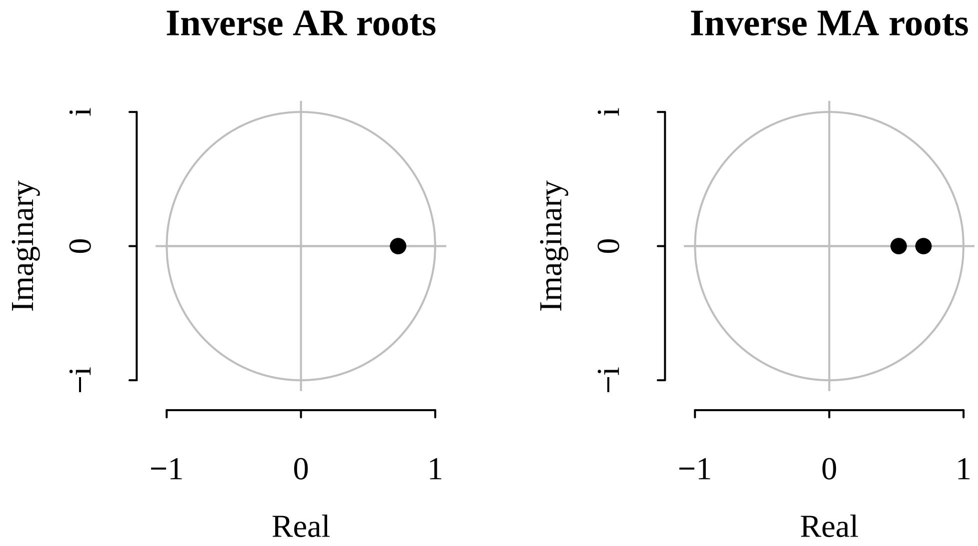
(d) (4 marks) Write out both models in equation form.

Forecasts from the ARMA(0,1) and ARMA(3,0) models for $h = 1, 2, \ldots, 8$ steps ahead are tabulated here:

h	ARMA(0,1)	ARMA(3,0)
1	2.993	3.065
2	3.014	3.052
3	3.014	2.952
4	3.014	3.025
5	3.014	3.016
6	3.014	3.020
7	3.014	3.008
8	3.014	3.014

(e) (5 marks) The two models have quite different functional forms and coefficient estimates, but quite similar forecasts. Give an explanation for this.

Referring to the table of AICc values, the ARMA(1,2) model may be another candidate model to try. The estimation of this model produces the following inverse roots.




```
Inverse MA roots: 0.722 0.517
Inverse AR root:  0.723
```

(f) **(7 marks)** Give an explanation of what information these two plots suggest about the specification of the ARMA(1,2) model.

If you run out of space for Question 3, use the extra pages at the end of the exam paper.

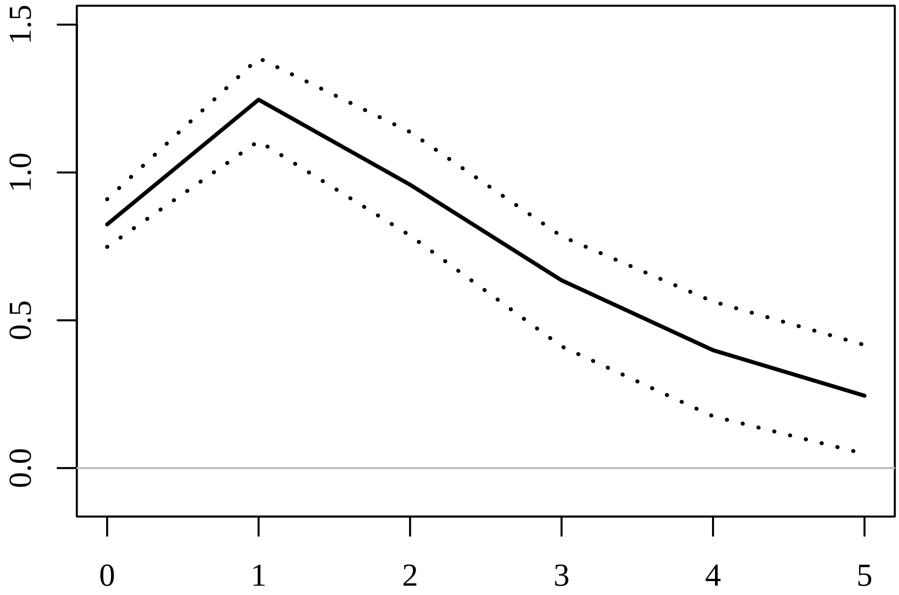
☐ **Please tick this box if you used extra pages for any part of Question 3.**

Question 4. (10 marks)

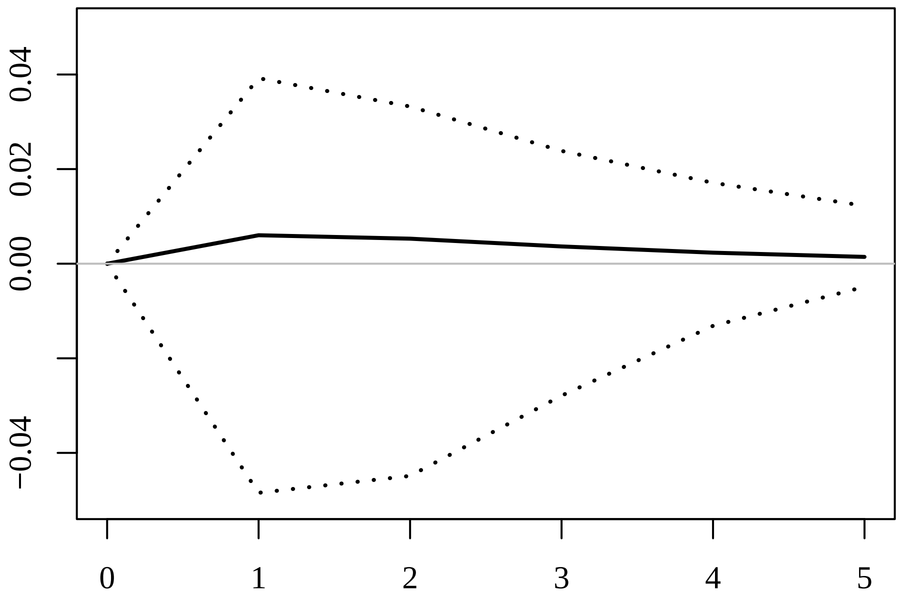
We have $n = 300$ observations on two time series, $Y_{1,t}$ and $Y_{2,t}$, to use as estimation sample, with the following $h = 5$ observations retained for forecast evaluation purposes. Following specification searches, bivariate VAR(1) model (`VAR`).

Using the VAR(1) model for $Y_{1,t}$ and $Y_{2,t}$, *orthogonalised* impulse response functions were calculated and plotted below, along with 95% confidence intervals (in dashed lines).

Response of $E(Y_{2,n+h}|\mathscr{Y}_n)$ to impulse to $Y_{1,n}$



Response of $E(Y_{1,n+h}|\mathscr{Y}_n)$ to impulse to $Y_{2,n}$



(a) **(10 marks)** Provide full interpretations of these two impulse response functions.

If you run out of space for Question 4, use the extra pages at the end of the exam paper. Please clearly indicate there the question number/part you are answering.

☐ **Please tick this box if you used extra pages for any part of Question 4.**

Question 5. (14 marks)

Consider a model for a quarterly time series Y_t:

$$Y_t = \beta_0 + \beta_1 Q_{1,t} + \beta_2 Q_{2,t} + \beta_3 Q_{3,t} + Z_t$$
$$Z_t = 0.3 Z_{t-1} + 0.4 Z_{t-2} + U_t + 0.5 U_{t-1},$$

where $Q_{1,t}$, $Q_{2,t}$ and $Q_{3,t}$ are seasonal dummy variables taking the value 1 in quarters 1, 2 and 3 respectively, and the value 0 otherwise.

(a) **(2 marks)** Express the equation for Z_t using lag operators and polynomials.

(b) **(4 marks)** Is it possible that the model for Z_t is stationary? Explain why or why not, and why the answer needs to be qualified by "is it possible".

(c) **(2 marks)** Is the model for Z_t is invertible? Explain why or why not.

(d) **(2 marks)** Does the model for Z_t contain a common factor? Explain why or why not.

(e) **(4 marks)** Is it possible that the entire model for Y_t (i.e. both equations considered together) is stationary? If necessary, explain any special case(s) that affect your general answer.

If you run out of space for Question 5, use the extra pages at the end of the exam paper. Please clearly indicate there the question number/part you are answering.

☐ **Please tick this box if you used extra pages for any part of Question 5.**

Question 6. (16 marks)

Let Y_t be a time series that satisfies the following AR(1)-ARCH(1) model

$$\begin{aligned} &Y_t = \beta_0 + Z_t \\ &Z_t = \phi_1 Z_{t-1} + U_t \\ &E(\, U_t \,|\, \mathscr{Y}_{t-1}\,) = 0 \\ &E(\, U_t^2 \,|\, \mathscr{Y}_{t-1}\,) = \omega + \alpha_1 U_{t-1}^2. \end{aligned}$$

(a) **(2 marks)** Is it valid to consider U_t as the *one-step-ahead prediction error* in this model when it has the ARCH(1) structure?

(b) **(2 marks)** Why is it required to impose $\omega > 0$ and $\alpha_1 > 0$ in this model?

(c) (6 marks) What parameter restriction is required for U_t to be stationary? Under this restriction, derive the *unconditional* mean and variance of U_t.

(d) **(4 marks)** Give expressions for the conditional mean and conditional variance of Y_t, i.e. $E(\, Y_t \,|\, \mathscr{Y}_{t-1} \,)$ and $\operatorname{var}(\, Y_t \,|\, \mathscr{Y}_{t-1} \,)$, *as functions of lags of* Y_t.

(e) **(2 marks)** How would such expressions for $E(\, Y_t \,|\, \mathscr{Y}_{t-1} \,)$ and $\operatorname{var}(\, Y_t \,|\, \mathscr{Y}_{t-1} \,)$ be useful for constructing a normal 95% prediction interval for the one-step-ahead forecast $E(\, Y_{n+1} \,|\, \mathscr{Y}_n \,)$?

(f) **(2 marks)** Referring to your answer to (d), which lags of Y_n are used in computing the prediction interval in the AR(1)-ARCH(1) model in this question?

(g) **(2 marks)** Referring to your answer to (d), how does value of the one-step-ahead forecast for Y_{n+1} relate to the value of the ARCH(1) coefficient (α_1)?

(h) **(2 marks)** Referring to your answer to (d), how does the width of the one-step-ahead prediction interval for Y_{n+1} relate to the value of the ARCH(1) coefficient (α_1)?

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